Data Science
Foundational Level Training

Presented By : Harsh Dalal

We Make You Trust Data

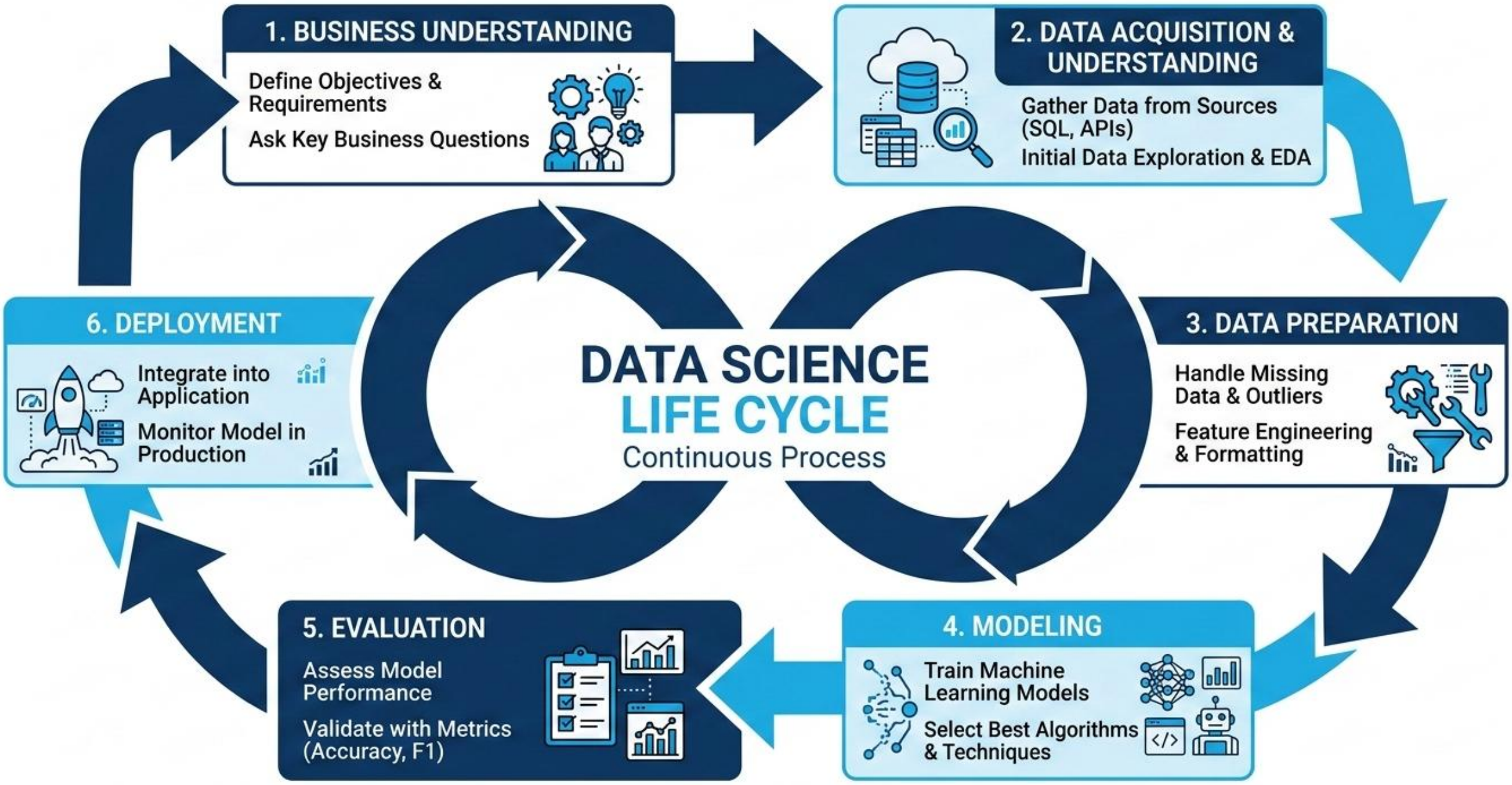
Trainer Introduction

Harsh Dalal – AI Trainer & Consultant

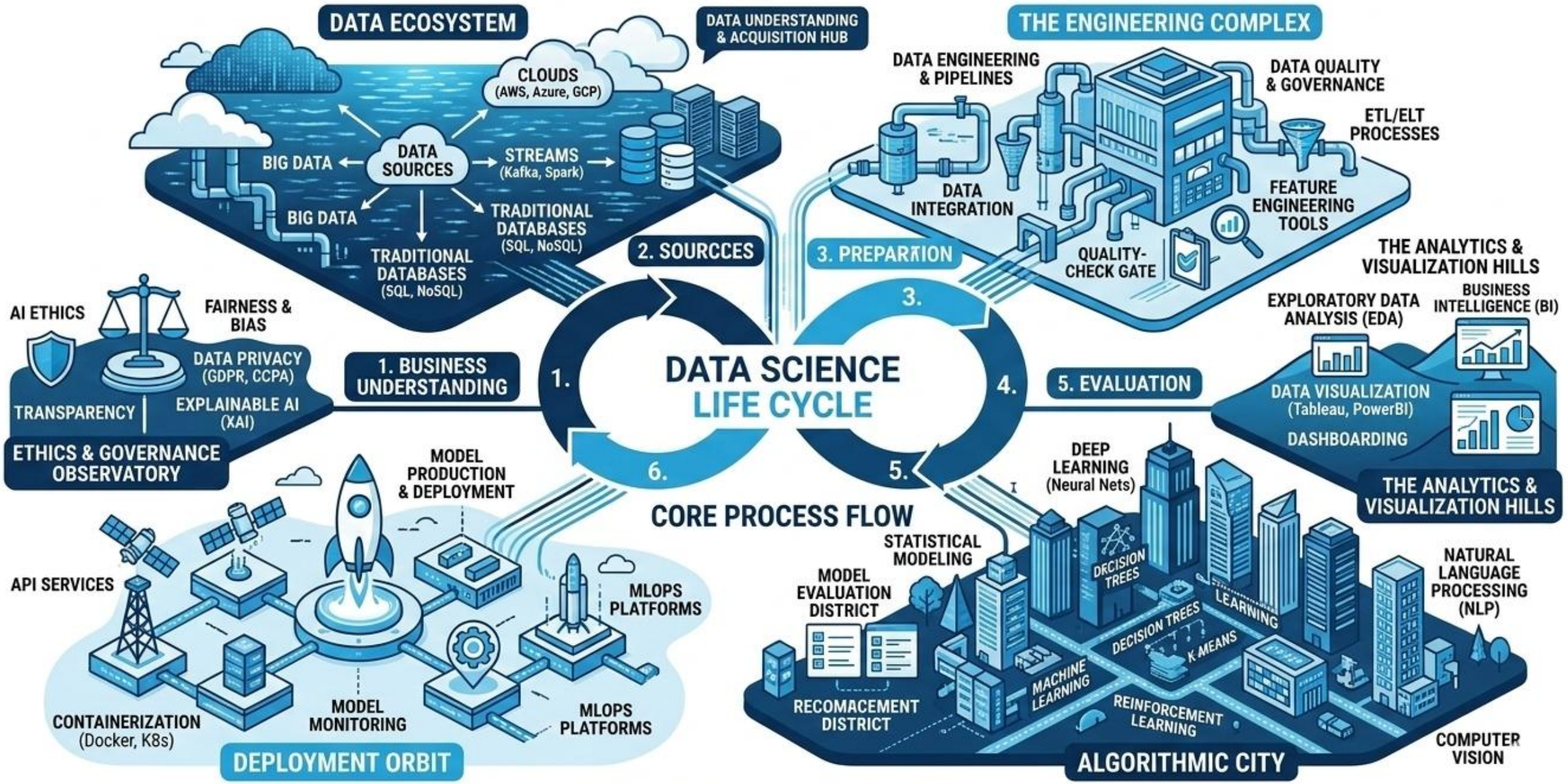
- Expertise in Generative AI, Agentic AI, RAG, and LLM Applications
- 10+ years of experience across industry and academic training
- 250+ certifications and achievements in Data, Cloud and AI technologies across Google Cloud Platform (GCP), Microsoft Azure, and AWS

Agenda

- Introduction to Data Science, its evolution, and how it differs from Data Analytics and Business Intelligence.
- Understanding core components of Data Science including statistics, programming, and domain knowledge.
- Exploring different types of data and reviewing enterprise data examples from finance, healthcare, and e-commerce.
- Learning the Data Science lifecycle and understanding key roles such as Data Analyst, Data Scientist, and Data Engineer.
- Hands-on activities including dataset exploration, business problem mapping, and lifecycle-based case study exercises.

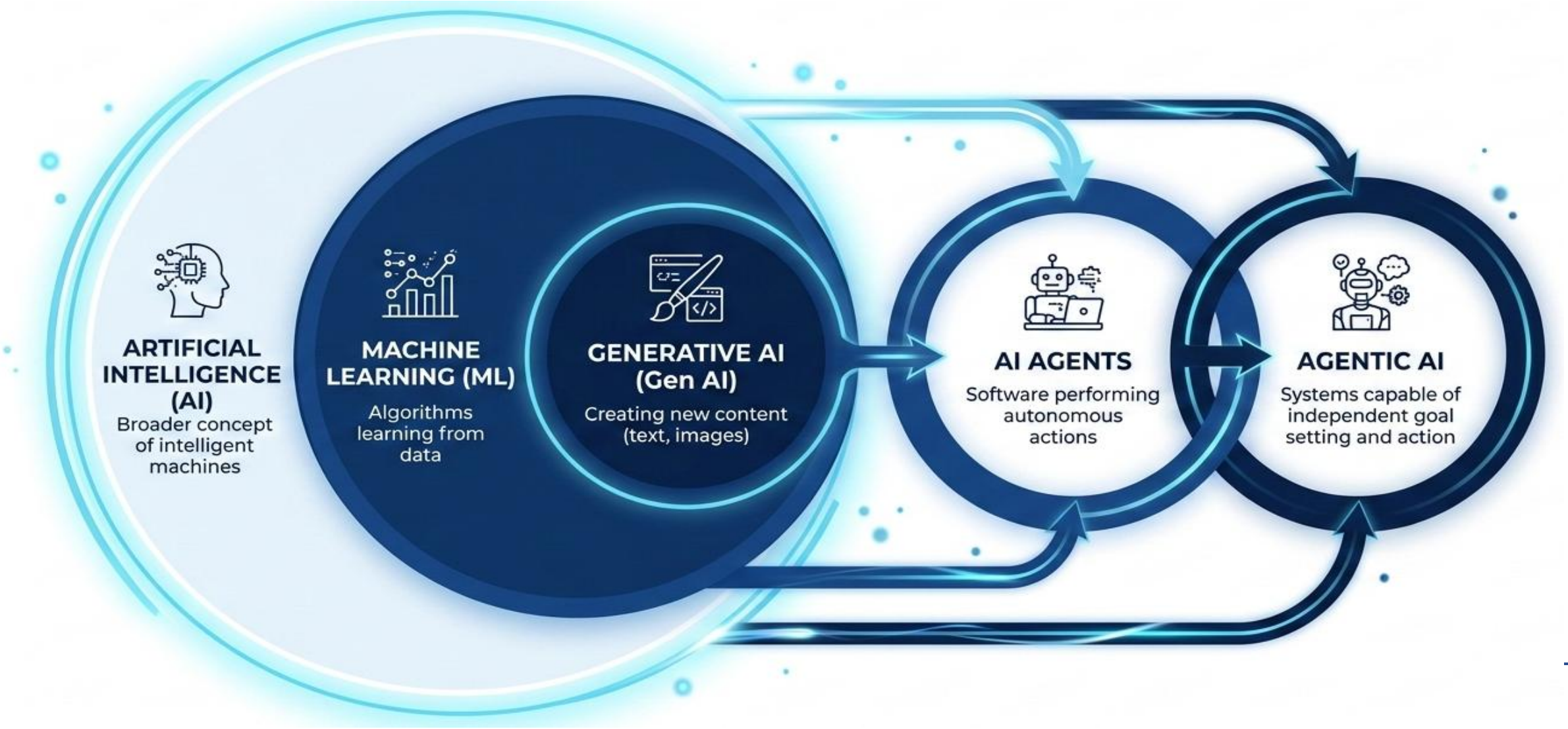

1. BUSINESS UNDERSTANDING
Define Objectives & Requirements
Ask Key Business Questions
2. DATA ACQUISITION & UNDERSTANDING
Gather Data from Sources (SQL, APIs)
Initial Data Exploration & EDA
3. DATA PREPARATION
Handle Missing Data & Outliers
Feature Engineering & Formatting
4. MODELING
Train Machine Learning Models
Select Best Algorithms & Techniques
5. EVALUATION
Assess Model Performance
Validate with Metrics (Accuracy, F1)
6. DEPLOYMENT
Integrate into Application
Monitor Model in Production
DATA SCIENCE
LIFE CYCLE
Continuous Process

THE COMPREHENSIVE LANDSCAPE OF DATA SCIENCE

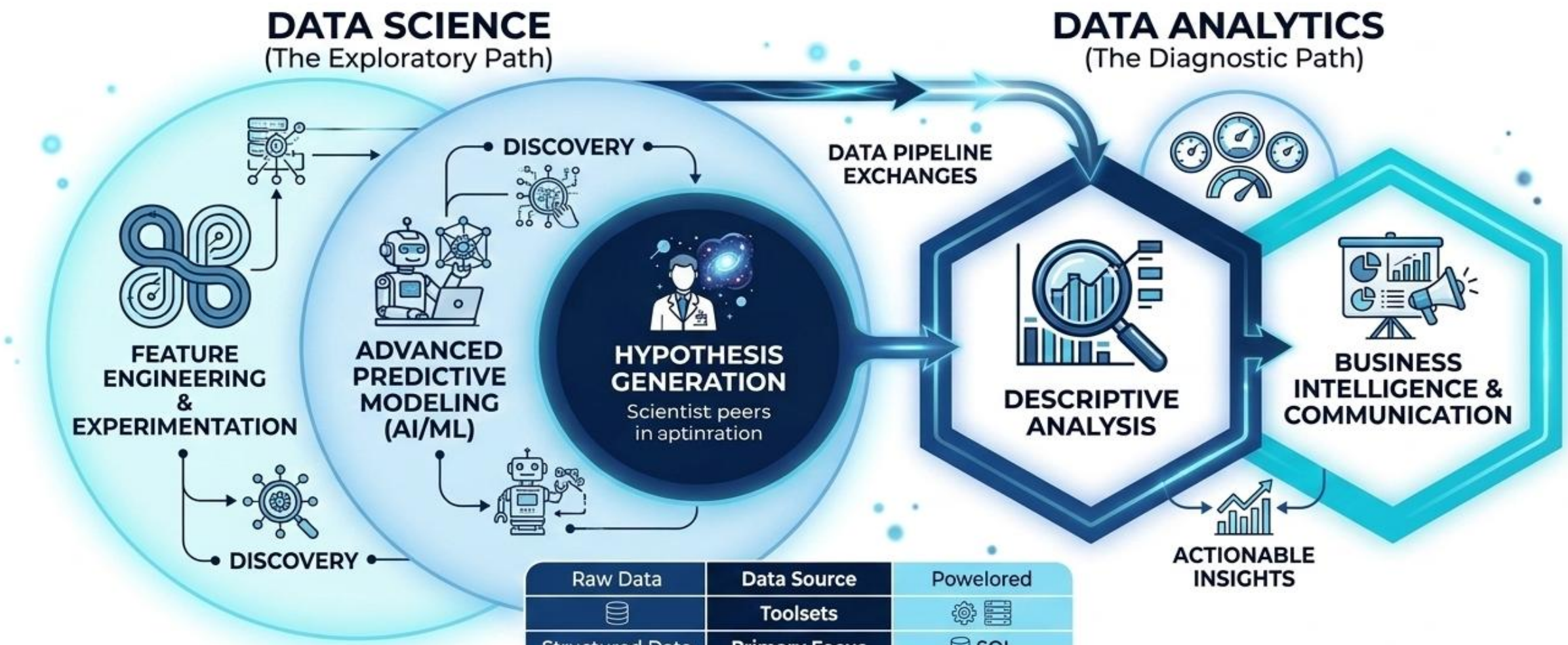


A holistic view of domains, tools, and processes.

AI Landscape



Data Science vs Data Analytics

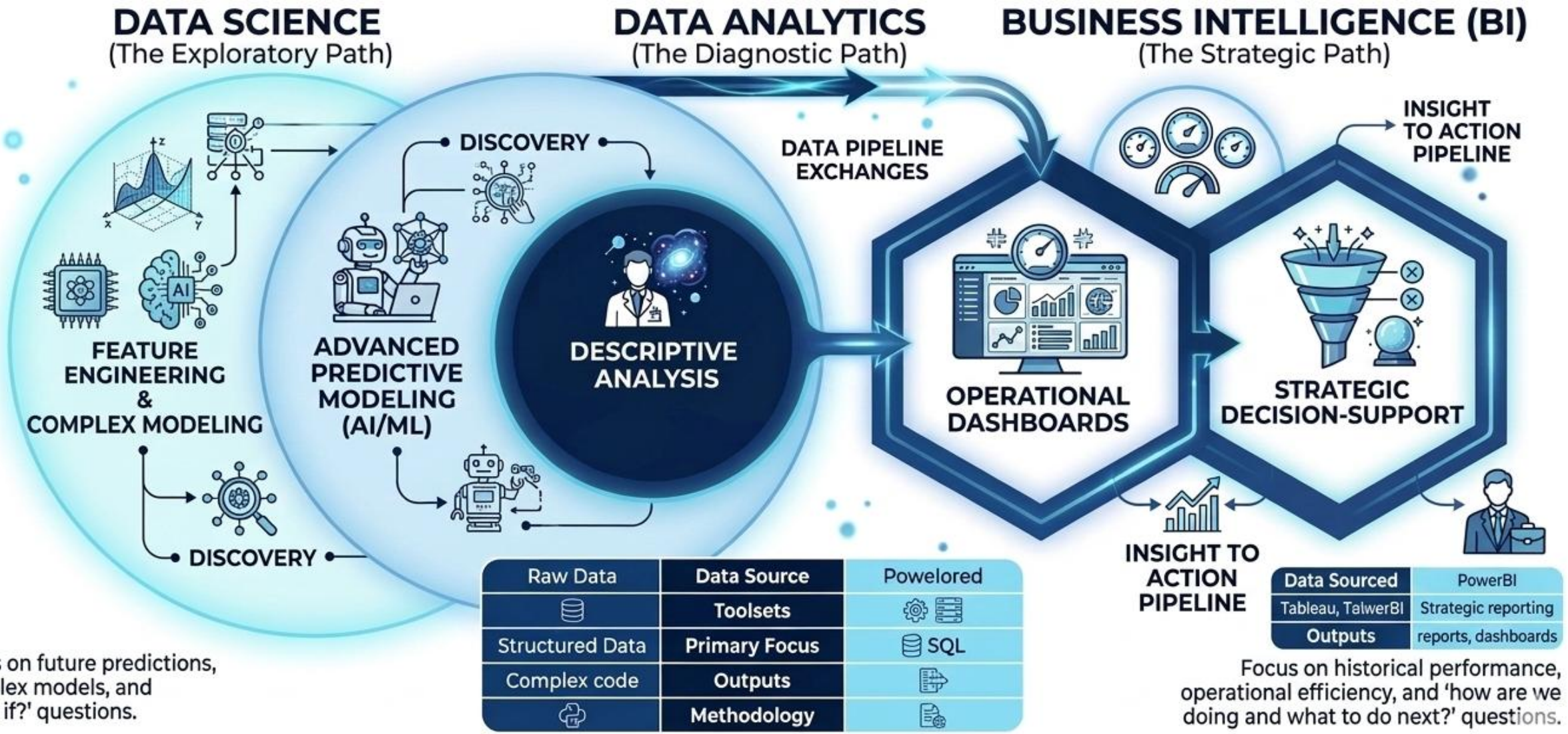


Raw Data	Data Source	Powelored
	Toolsets	
Structured Data	Primary Focus	SQL
Complex code	Outputs	
	Methodology	

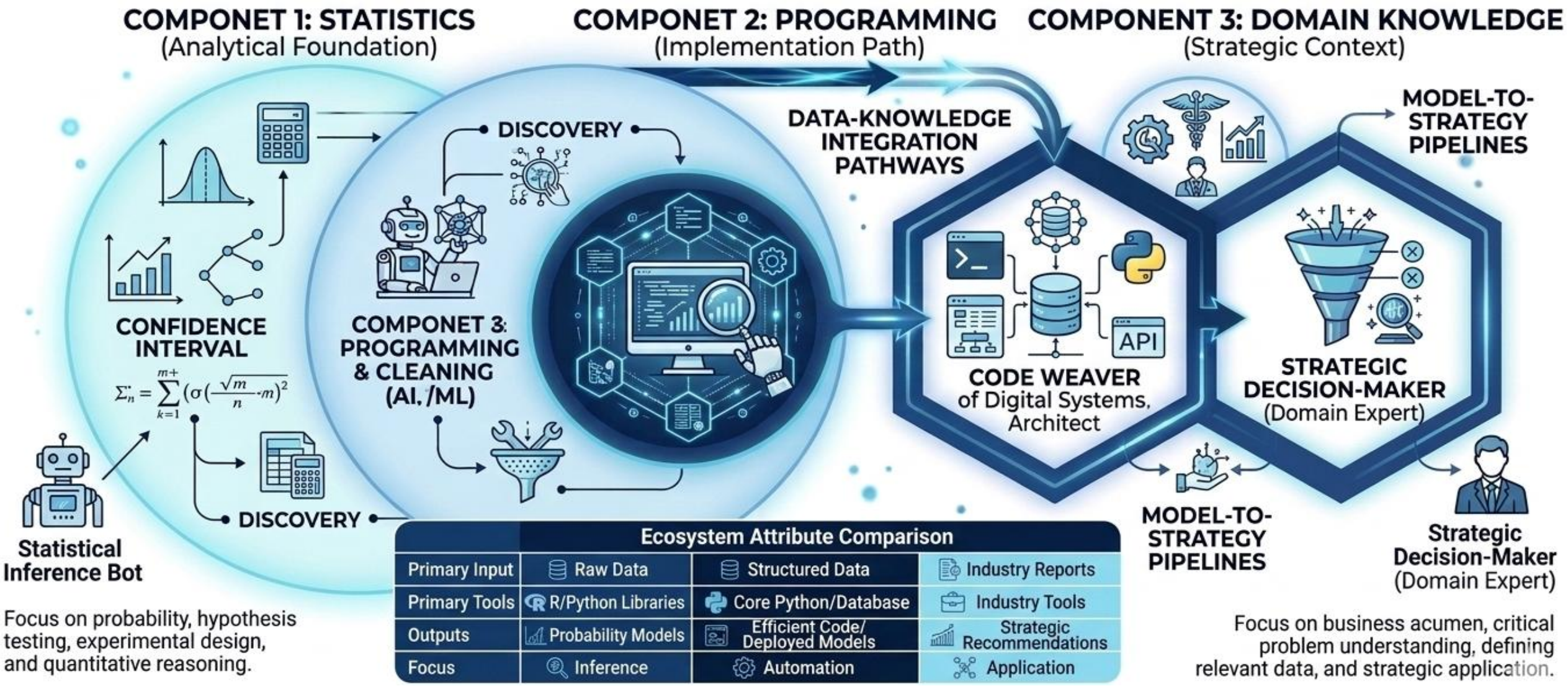
Focus on future predictions, creating new models, and answering 'what if?' questions.

Focus on understanding historical data, identifying patterns, and answering 'what happened and why?' questions.

Data Science vs Data Analytics vs Business Intelligence

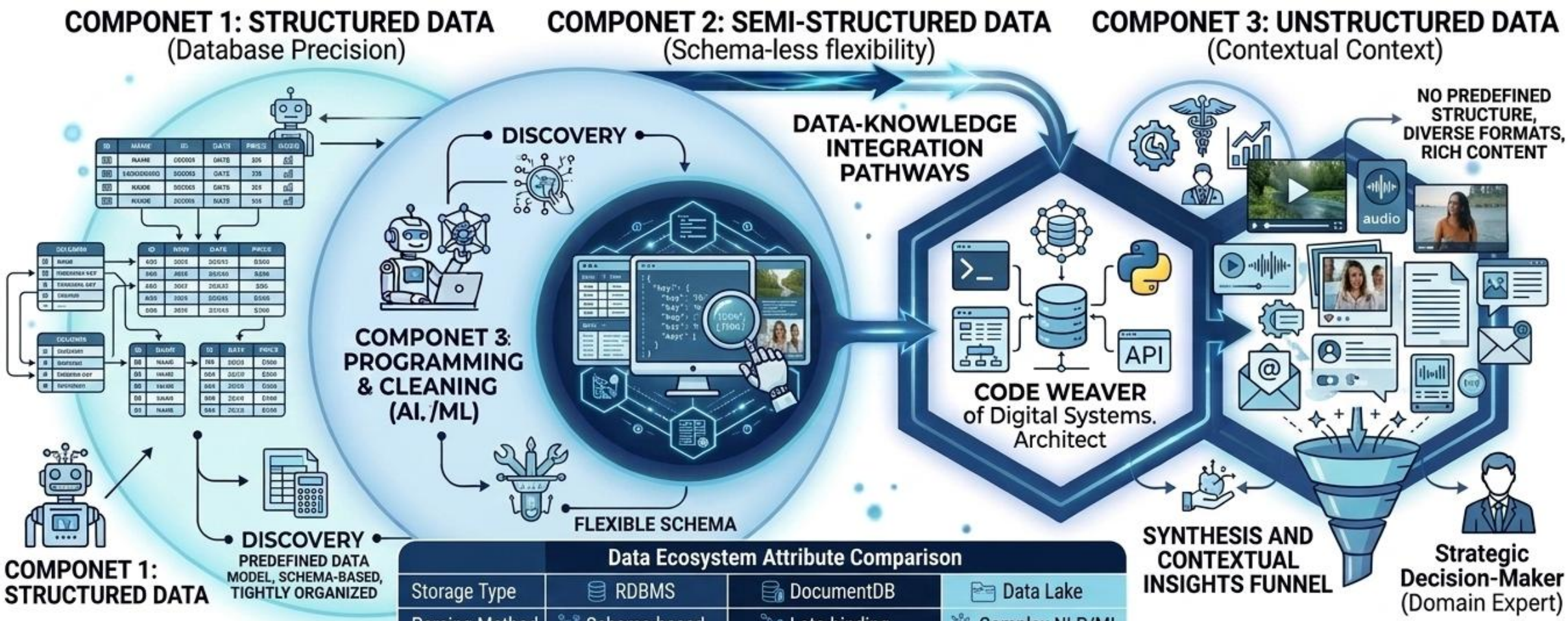


Interdisciplinary components



Ecosystem Attribute Comparison			
Primary Input	Raw Data	Structured Data	Industry Reports
Primary Tools	R/Python Libraries	Core Python/Database	Industry Tools
Outputs	Probability Models	Efficient Code/ Deployed Models	Strategic Recommendations
Focus	Inference	Automation	Application

Types of Data: Structured, Semi-Structured, and Unstructured



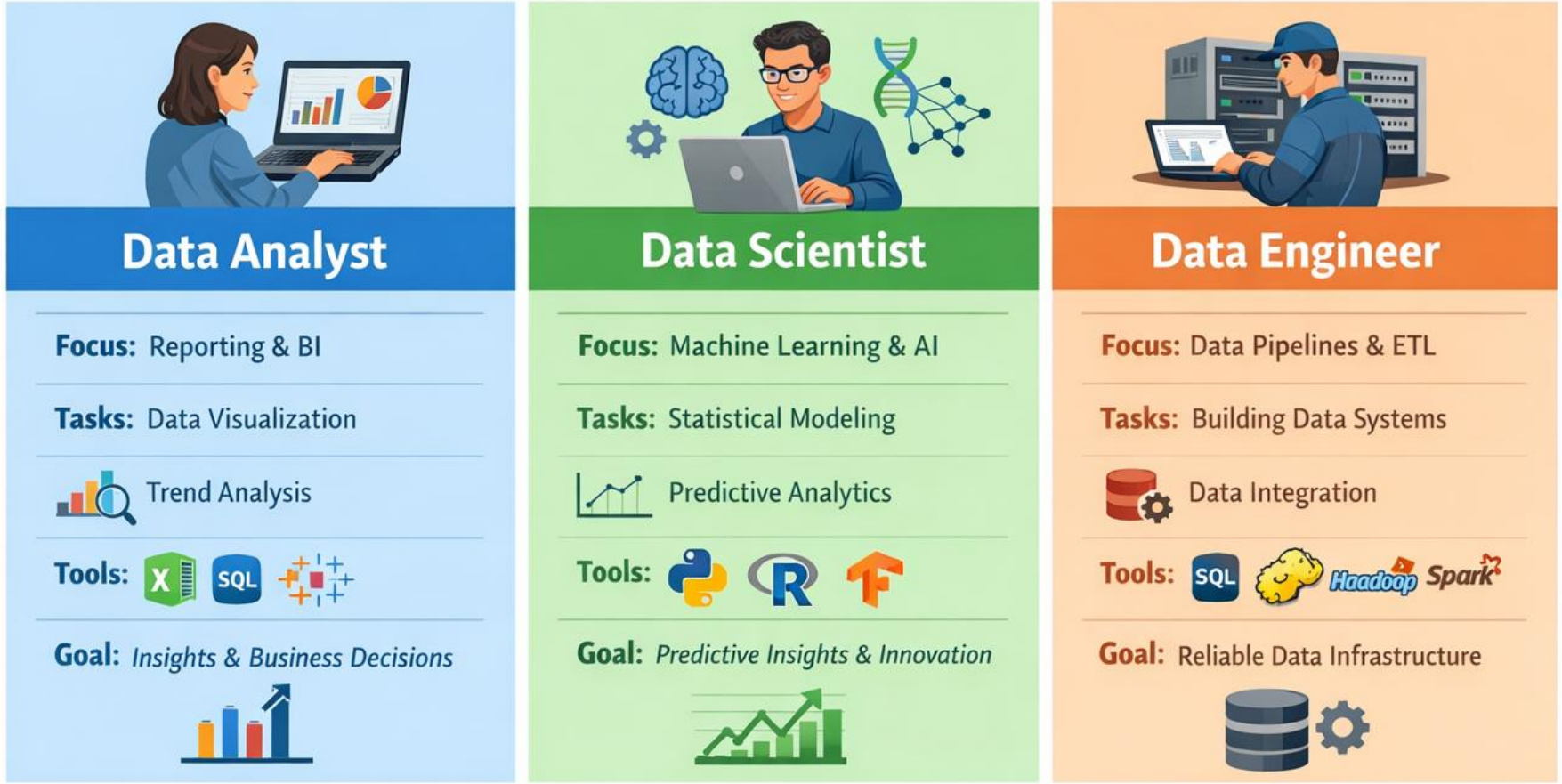
Data Ecosystem Attribute Comparison			
Storage Type	RDBMS	DocumentDB	Data Lake
Parsing Method	Schema-based	Late-binding	Complex NLP/ML
Typical Uses	Transactions	Content-heavy systems	Media Analysis
Primary Tools	SQL	NoSQL	NLP/CV

Focus on relational databases, tabular format, specific field definitions, and SQL queries.

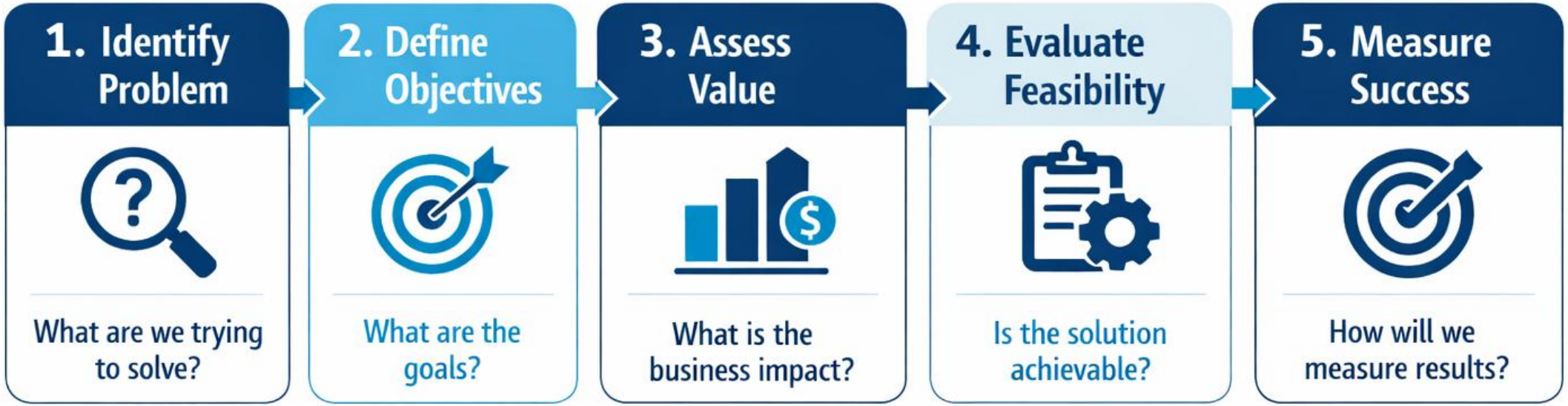
Focus on text, video, audio, imagery, complex human-generated content, and natural language processing (NLP).

Enterprise data examples from finance, healthcare, and e-commerce

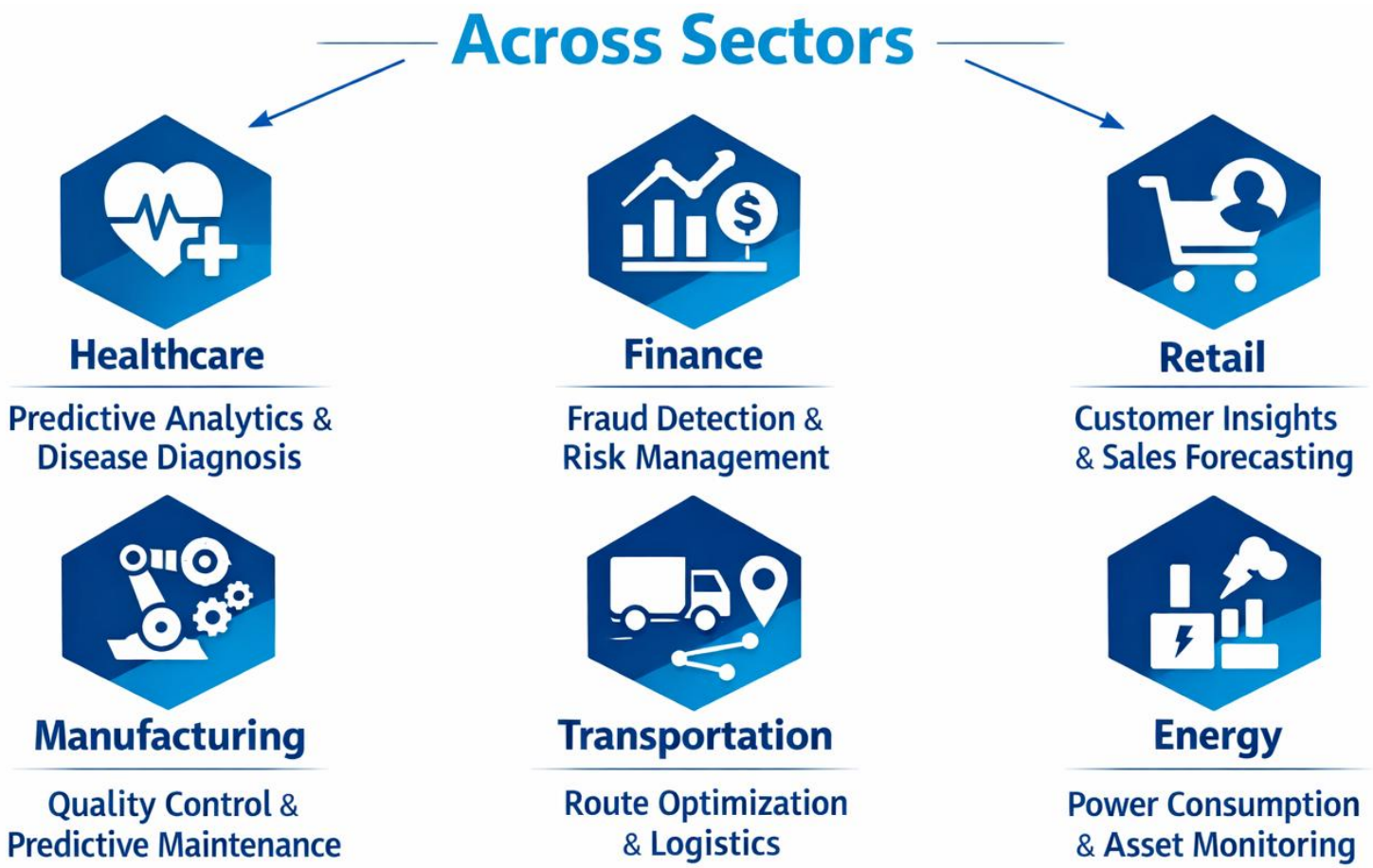
Data Analyst vs Data Scientist vs Data Engineer



Business problem framing in Data Science projects

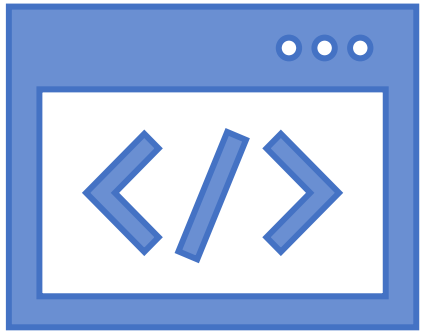


Industry use cases of Data Science



Summary

- Students gained a clear understanding of the concept, evolution, and scope of Data Science.
- Participants learned the differences between Data Science, Data Analytics, and Business Intelligence.
- Learners explored different data types and examined enterprise data examples across industries.
- The session clarified the Data Science lifecycle and the responsibilities of different data-related roles.
- Practical exercises helped students translate business questions into data analysis problems and analyze sample datasets.



Practical Activity

Group of 3- 4 Students

Per Activity 5-10 Minutes

We Make You Trust Data

Activity 1 : Mapping Business Problems to Data Sources

- **business problem scenario**
  - **"An e-commerce company wants to reduce cart abandonment"**
- **Identify possible data sources** such as:
  - Website clickstream logs
  - …
  - …
- create a **mapping table**:

Business Question	Possible Data Source
Why are users abandoning carts?	Website clickstream data
……	……

Activity 2. Identifying the Correct Stage of the Data Science Lifecycle

- scenarios like:
  - “Cleaning missing values from a dataset”
  - “Collecting customer transaction data”
  - “Building a prediction model”
- classify them into lifecycle stages:
  - Data Collection
  - Data Cleaning
  - Data Analysis
  - Modeling
  - Deployment

Activity 3 : Exploring Different Types of Data

Data Type	Example
Structured	CSV file with sales data
Semi-Structured	JSON API response
Unstructured	Customer reviews text

1.Structured Data – Columns & Schema

Order_ID	Customer_ID	Product_Name	Quantity	Price	Order_Date
1001	C101	Laptop	1	65000	2025-01-10
1002	C205	Smartphone	2	25000	2025-01-11
1003	C309	Headphones	1	3000	2025-01-12

2. Semi-Structured Data – Key-Value Hierarchy (JSON Example)

JSON

```
{
  "order_id": 1001,
  "customer": {
    "customer_id": "C101",
    "name": "Rahul Mehta",
    "city": "Surat"
  },
  "product": {
    "product_name": "Laptop",
    "category": "Electronics",
    "price": 65000
  },
  "quantity": 1,
  "payment": {
    "method": "Credit Card",
    "status": "Completed"
  }
}
```

3. Unstructured Data – Free Text (Customer Reviews)

Review_ID	Customer_Review
R001	"The laptop performance is excellent and battery life is very good."
R002	"Delivery was late but the product quality is decent."
R003	"Very poor customer support, I had issues with installation."
R004	"Great value for money and fast delivery. Highly recommended."

Activity 4 : Case Study: Selecting Data Science Roles

- **project scenario**:
  - “Healthcare hospital wants to predict patient readmission.”
- Students assign roles:

Task	Role
Collect patient records	
Analyze patterns	
Build prediction model	

Activity 5. Converting Business Question into Data Problem

- business question:
  - “Why are sales decreasing in the last quarter?”
- convert it into **data questions**:
  - Which products show sales decline?
  - ……
  - ……
- Then define:
  - Required dataset
  - Metrics to analyze
  - ……..

Activity 6 : Lifecycle Mapping Exercise

- Give a **complete real-world scenario**:
  - “Bank wants to detect fraudulent transactions.”
- Students map steps:
  - Data Collection →
  - Data Cleaning →
  - Feature Engineering →
  - Model Training →
  - Deployment →
- Students create a **lifecycle flow diagram**.

Activity 7 : Kaggle Exploration

kaggle Competitions Datasets Models Benchmarks Code Discussions Blog ••• Search Sign In Register

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Activity 8 : UCI Machine Learning Repository

UC Irvine Machine Learning Repository

Datasets | Contribute Dataset | About Us | Search datasets... | Login

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Welcome to the UC Irvine Machine Learning Repository

We currently maintain 689 datasets as a service to the machine learning community. Here, you can donate and find datasets used by millions of people all around the world!

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Regression | 3.06M Instances | 2 Features

Any Query

Thank You

Happy learning